Portfolio Status

5 August 2026

In the twenty-four hours since the last edition the portfolio gained 30 signed verdicts, took delivery of a founder directive that is now bound to the loop by content hash, and had three of its never-attacked modules meet an adversary for the first time. Two of those three were refuted.

This document reports what the evidence supports and, in each section, what it does not.

THE FINDING THIS EDITION LEADS WITH. A signing key that is **not in the trust store** – `local-9028773f49591193` – carries 10 signature blocks in the evidence tree, including 7 Phase-0 KPI files. Signed is not the same as anchored, and nothing until now measured the difference.

EVIDENCE BASE. 108 signed verdicts across 25 modules (68 refuted, 38 survived, 2 blocked) · 238 autonomous fires · 75 signed repair receipts · 83 Phase-0 KPI artifacts · loop config at 0569b517f649

No promotion, procurement, safety, synthesis or freedom-to-operate authority is asserted by this document. Every number is computed from the signed evidence tree on xbabe2; none is typed by hand.

EXECUTIVE SUMMARY

Where the portfolio actually stands

Evidence. 108 signed, independently verifiable verdicts across 25 of 41 modules. 68 refutations, 38 survivals. A refutation is a success: it is the system finding its own defect before a customer does.

WHAT MOVED SINCE 2026-08-04

- **Three modules met an adversary for the first time** – candidate-contract, cro-protocol-packager and envelope-gates had zero verdicts before today. Two of the three were refuted, each against a criterion the founder pre-committed *before* the attack ran.
- **A founder directive is now bound to the loop by content hash.** 46 obligations, each with an owner class; 48 campaign items cite the document by SHA-256; a coverage report reconciles the two and fails closed if any obligation loses its owner.
- **A non-anchored signing key was found in the Phase-0 KPI surface** – see the trust-chain section. This is new; no prior edition measured it.
- **The declared-vs-derived Phase-0 gap is unchanged at 8 modules.** It needs a human ceremony the loop is forbidden from performing.

WHAT THIS DOCUMENT CANNOT TELL YOU

- **No module has crossed Phase-1.** Phase-1 means a locked external benchmark passing under independent evaluation and review. Zero modules qualify, and no amount of further autonomous work can change that — it requires owned, candidate-bound wet-lab measurement.
- **The Phase-0 derivation quoted here is as-of 2026-08-04 (file mt** and predates today's 21 fires. It is quoted dated rather than re-run, because re-running it risks the exact defect that broke the bindings in the first place.
- **Survivals are weaker evidence than refutations.** A refutation exhibits a concrete failure. A survival only says an attack did not succeed — which is why every survival must now carry a positive control proving the attack could have fired.

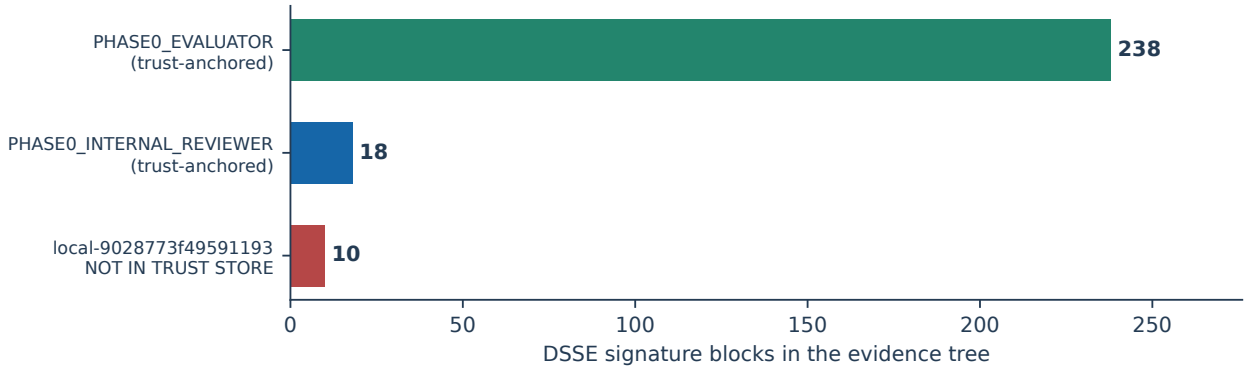
Red team. The most load-bearing number in this document is 108. It counts verdicts, not correctness. 68 of them are the system reporting its own defects, and only 38 are claims that withstood attack — and of those, the ones predating 2026-08-05 were not required to prove the attack could have failed. Read the survival count as a lower bound on work done, never as an upper bound on quality.

TRUST CHAIN

Signed is not the same as anchored

New this edition. A key that is **not in the trust store** signs 10 artifacts in the evidence tree. The trust store (`config/phase0-trust-store.v1.json`) anchors exactly two identities. This key is neither of them.

WHO SIGNS WHAT



The two anchored identities behave as designed. `PHASE0_EVALUATOR` signs 238 blocks; `PHASE0_INTERNAL_REVIEWER` signs 18 across 18 modules. **The dual-identity path is exercised, not theoretical** — every one of the 9 modules the verifier derives at Phase-0 carries an evaluator receipt, a reviewer receipt and exactly one movement event.

THE EXPOSURE

module	unanchored	total KPIs	share
sourceability-ehs	7	9	77%

The key is `local-9028773f49591193`. Its name marks it as a local development key, and the portfolio's own intake validator exists to reject exactly this: a receipt whose signature verifies arithmetically but whose key no trust store vouches for.

WHAT IT DOES AND DOES NOT MEAN

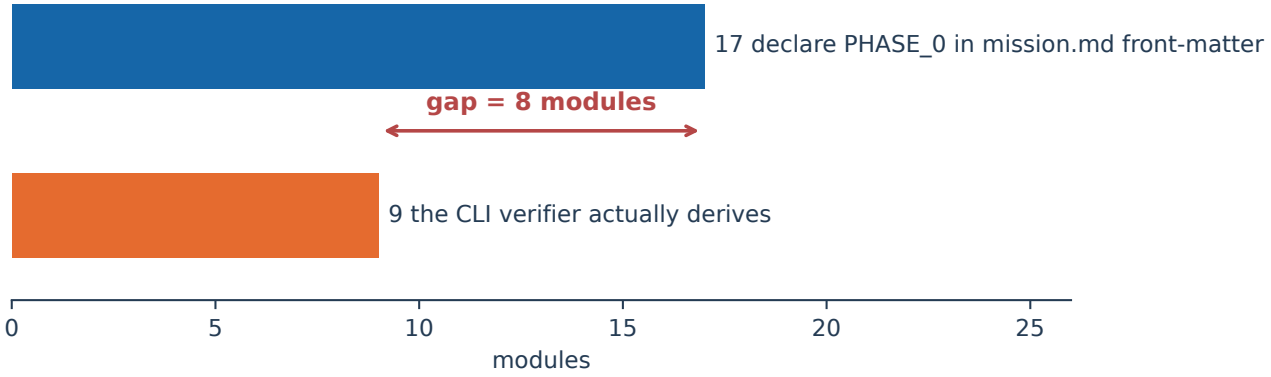
- It does **not** mean those artifacts are forged. The signatures are well-formed; the key is simply unvouched.
- It does mean any claim resting on them is **not trust-anchored**, so it cannot support a promotion, and it should not be quoted externally as Phase-0 evidence.
- None of the affected files is cited in that module's `mission.md`, which limits the blast radius — but nothing was *enforcing* that limit.

Red team. This section's own method has a hole worth naming. It counts signature blocks by key id; it does not re-verify every signature cryptographically against its payload. A block whose key id says `PHASE0_EVALUATOR` but whose signature does not actually verify would be counted as anchored here. The loop's per-fire verifier does perform real Ed25519 verification over the PAE encoding, so newly-credited artifacts are checked — but this table is an inventory, not a proof.

PHASE-0

17 declare it. 9 derive it.

All 8 fail the same gate: EVALUATION_RESULT_MISMATCH.
Signatures are valid; the evidence-byte binding is broken.



The quotable number is 9. Not 17. A module's front-matter is a declaration; the CLI verifier's derivation is a reconstruction from signed movement events. Where they disagree, the derivation is authoritative and the declaration is aspiration.

WHY THE GAP EXISTS

All 8 gap modules fail one gate: `EVALUATION_RESULT_MISMATCH`. A test harness once wrote `evidence/phase0/benchmark/result.v1.json` directly, while the evaluator had hash-pinned those exact bytes into an immutable signed receipt. The harness run destroyed evidence the receipt still points at. The signatures remain valid; the binding does not. Original bytes are unrecoverable — absent from git, and the oldest archive already holds harness-shape bytes, so the overwrite predates every snapshot.

- `cro-protocol-packager`
- `evidence-intake-validator`
- `known-answer-benchmarks`
- `molecule-design-gate`
- `optimizer-orchestrator`
- `redteam-audit`
- `report-packager`
- `source-ledger`

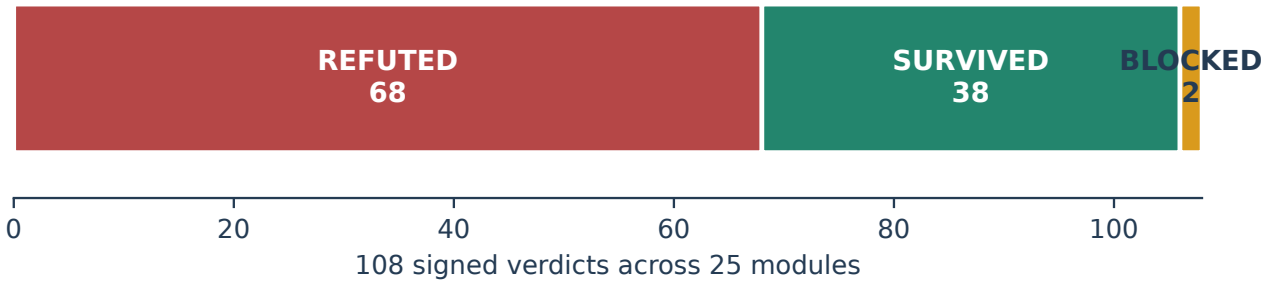
WHAT CLOSES IT

A governed re-evaluation tool exists and has been tested, but has **never been run**: it requires two distinct, non-revoked, non-development identities and refuses a single signature or a repository-local key by design. That is a human ceremony. It restores **eight** of the nine — `molecule-property` has no independent-review receipt and 0 movement events, so the reconcile branch cannot restore it; it needs a full re-evaluation plus measured data.

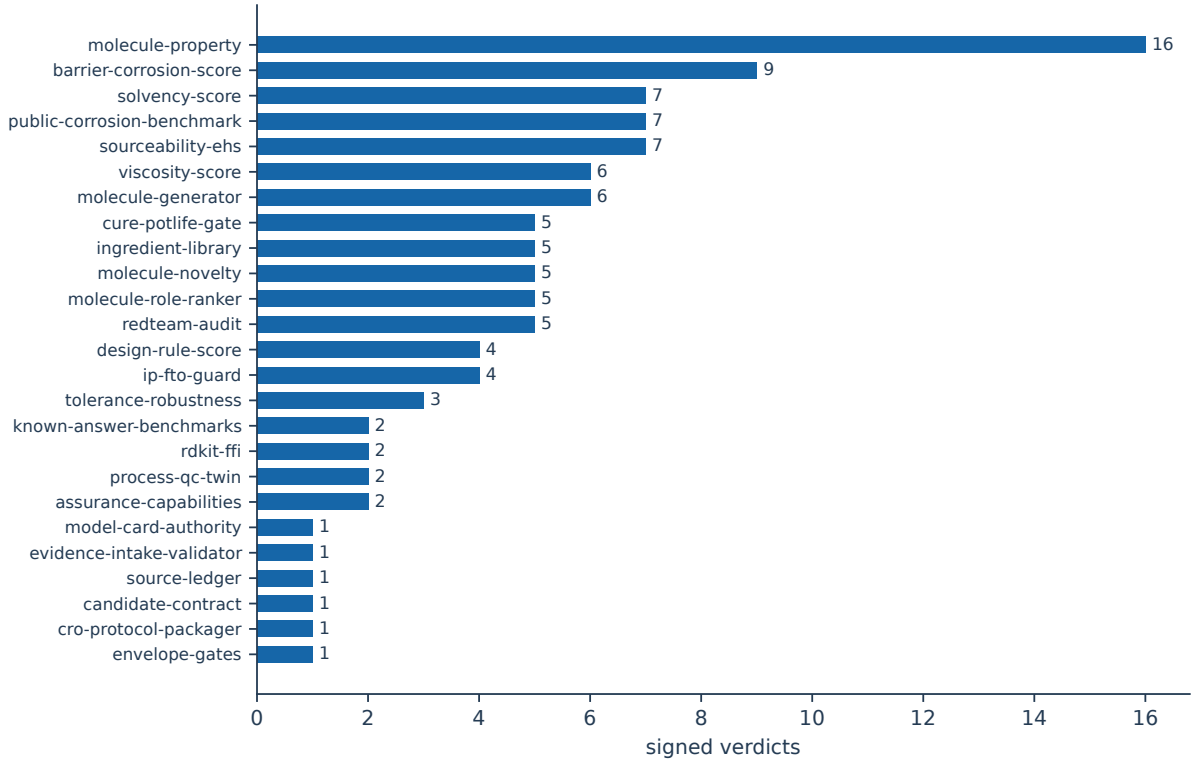
Red team. Do not let a summary round eight up to nine. And note the number above is dated: the decision report was generated 2026-08-04 (file mt, before today's fires. Three modules gained their first verdicts today; none of them changes a Phase-0 derivation, but this document cannot prove that without re-running the verifier, and re-running it carries the risk that caused the gap.

WHERE THE ATTACKS LANDED

108 verdicts, 25 modules



Verdict coverage is uneven by design: directed items go where the evidence is thinnest.



EVERY FIRE OF 2026-08-05

UTC	item	module	verdict	time
00:00	V-B02	barrier-corrosion-score	REFUTED	16m
01:00	V-B03	solvency-score	SURVIVED	7m
02:00	R-Z99	redteam-audit	SURVIVED	8m
03:00	W-QC	process-qc-twin	SURVIVED	6m
04:00	W-NOV1	molecule-novelty	SURVIVED	8m
05:00	W-TOL1	tolerance-robustness	REFUTED	9m
06:00	W-GEN1	molecule-generator	REFUTED	6m
07:00	W-FTO1	ip-fto-guard	REFUTED	11m
08:00	W-EHS1	sourceability-ehs	REFUTED	9m
09:00	W-VISC1	viscosity-score	REFUTED	17m
10:00	W-PROP1	molecule-property	SURVIVED	7m
11:00	W-RTA1	redteam-audit	REFUTED	7m
12:00	W-RTA2	redteam-audit	REFUTED	10m
14:00	V-A10	solvency-score	SURVIVED	5m
15:00	V-A06v2	molecule-generator	SURVIVED	6m
16:00	W-ASSUR1	assurance-capabilities	SURVIVED	24m
17:00	W-ING1	ingredient-library	SURVIVED	8m
18:00	W-FFI1	rdkit-ffi	REFUTED	17m
19:00	T-01	candidate-contract	SURVIVED	24m
20:00	T-02	cro-protocol-packager	REFUTED	13m
21:00	T-03	envelope-gates	REFUTED	11m

21 fires. The gap at 13:00 is not an idle hour — it is a fire whose work completed and was never recorded, because of a defect in the launcher introduced that afternoon. The operations document tells that story in full.

Red team. Verdict counts measure activity, not coverage. 25 modules carry a verdict; the rest carry none, and a module with no verdict is not 'passing' — it is untested. Per-module counts are also incomparable: sixteen verdicts against `molecule-property` reflect where directed work was aimed, not that it is sixteen times better understood than a module with one.

THE FOUNDER DIRECTIVE

Bound to the loop, by hash

On 5 August at 17:46 a founder directive arrived as a Word document. By 18:35 it was rendered, indexed, placed in the evidence tree, and every relevant campaign item cited it by SHA-256. By 21:12 three of its probes had run and two had refuted their targets.

Binding. 46 obligations (25 loop, 10 human, 11 lab). 48 campaign items carry a `directive_ref`. Document hash `d cedb048e1142529de84aa6d741a065f…` — verified identical on both hosts and re-hashed by the launcher at every fire.

WHAT THE DIRECTIVE BOUGHT, MEASURED

- **cro-protocol-packager REFUTED.** A packet-authored decision threshold, paired with a format-valid but content-underived lock hash, passes validate, sign and verify end-to-end. The lock is checked for shape, never recomputed from what it locks. A CRO packet is meant to be a structured question; one that can set its own acceptance threshold can answer itself.
- **envelope-gates REFUTED.** An 18-cell metamorphic boundary grid found five pass-on-invalid-input slips — a value-free $\pm\infty$ bypass on a real registered policy, NaN-feature masking through Python's non-total-order `min()`, and no policy-provenance checking at all.
- **candidate-contract SURVIVED, and earned it.** All eight injection encodings were rejected; the typed struct was byte-identical to the clean control across every vector; and the fire carried a positive control proving its own detector fires when the defect is present.

Why this matters more than the count. Each of those three verdicts was measured against a refutation criterion the founder wrote down *before* the attack existed. That is the difference between a test and a demonstration. Two of three pre-committed criteria were met by real defects — in modules that had carried a composite score for weeks without ever having met an adversary.

COVERAGE, HONESTLY

6 of 46 obligations are closed: 6 of 25 loop-owned, 0 of 10 human-owned, 0 of 11 lab-owned. The last two rows cannot move without a human: the loop is structurally barred from executing them, and a guard in the scheduler now refuses to dispatch one regardless of its status.

Red team. A closed obligation here means its campaign items reached `done`, which means a signed receipt matched a glob. It does not mean an independent reader agreed the founder's intent was served. An audit item exists to check exactly that and has not yet run — until it does, treat 6 as an upper bound.

MODULE DEEP DIVES

Every module that carries a verdict

Ordered by verdict count. 25 of 41 modules carry at least one signed verdict; the remainder are untested, which is not the same as passing.

molecule-property

PRE-PHASE-0 UNVALIDATED · 16 verdicts (13 refuted / 3 survived) · 21 KPIs, 7 repair receipts · queue: 4 pending / 7 done / 3 blocked

Most recent verdict *(2026-08-05T10:07, SURVIVED)* W-PROP1: derived a per-group training-support threshold (T=125) directly from the ridge-fit KPI own sign-admissibility boundary (not assumed 500), partitioned 24 groups into 15 in-scope / 9 excluded (incl both W8 sign-flips), re-ran the original W8 pre-registered refute criterion restricted to the...

First verdict *(2026-08-01T05:13, REFUTED)* C7: cross-corpus external-validity (OpenPoly Tg RF <-> Polyuniverse Epoxy MD Tg, both directions) + leave-polymer-class-out proxy for density (RadonPy PI1070, no 2nd density corpus exists). Tg REFUTED (6.66x/16.60x...

Limits: no independent-review receipt; no movement event.

barrier-corrosion-score

PRE-PHASE-0 UNVALIDATED · 9 verdicts (4 refuted / 4 survived) · 3 KPIs, 3 repair receipts · queue: 1 pending / 5 done / 4 blocked

Most recent verdict *(2026-08-05T00:16, REFUTED)* adversarial edge case NOT in R-B02 tests: append duplicate of highest-frequency point to end of well-formed descending spectrum, calling barrier_index_from_spectrum() directly (bypassing validate_conditions) to test the documented fail-closed-on-its-own guarantee

First verdict *(2026-07-31T22:06, BLOCKED)* rights-package-assembly: hash-bound RIGHTS-REVIEW-REQUEST.md + consolidated data-acquisition-queue entry for USBR ST-19168 (catalog 4640/item 11562, sha 9e34fccc..., 10 coating systems) referencing the two...

Limits: no independent-review receipt; no movement event.

public-corrosion-benchmark

PRE-PHASE-0 UNVALIDATED · 7 verdicts (5 refuted / 2 survived) · 3 KPIs, 8 repair receipts · queue: 1 pending / 4 done / 1 blocked

Most recent verdict *(2026-08-04T08:07, SURVIVED)* independent verify: re-ran committed unmodified run_fold_significance.py (sha256 d6300bbd..., two-replay byte-identical, matches R-A08 embedded reattack stdout sha256 d7b154c6...); enumerated evidence/audit/ (8/8 on-disk receipts indexed in v2 KPI, sha256-matched, 0 missing/extra); real...

First verdict *(2026-08-01T23:04, REFUTED)* completeness audit of KPI surface vs evidence/audit/ tree: checked whether the 4 signed coating-side audit verdicts (Brasher-Kingsbury H0_REFUTED 6/6 + T0 data-integrity erratum, EIS discrimination-power n~90 required,...

Limits: no independent-review receipt; no movement event.

solvency-score

PHASE-0 LOCKED-INTERNAL · 7 verdicts (2 refuted / 5 survived) · 8 KPIs, 6 repair receipts · queue: 2 pending / 5 done / 2 blocked

Most recent verdict *(2026-08-05T14:04, SURVIVED)* second-pass independent re-verify: freshly-written harness re-derives all 4 R-A10 acceptance items from live sources (front-matter byte-identity, prose/front-matter consistency, W7 disclosure numbers vs live W7 receipt, BigSolDB headline vs live W6 receipt); NEW real Ed25519 cryptographic...

First verdict *(2026-07-31T04:30, REFUTED)* independent re-derivation of KPI->receipt_path->DSSE-Ed25519-signature->receipt_sha256 chain for all 4 flagged KPIs, cryptographic signature verify against pinned pubkey, byte-diff of KPI vs parent-receipt schema fields

Limits: evaluator and reviewer receipts present, movement event recorded, work queued.

sourceability-ehs

PRE-PHASE-0 UNVALIDATED · 7 verdicts (5 refuted / 2 survived) · 9 KPIs, 5 repair receipts · queue: 3 pending / 3 done / 0 blocked

Most recent verdict *(2026-08-05T08:09, REFUTED)* W-EHS1 live-source reconciliation: fresh independent fetch of PubChem PUG-View aggregate (CID 2955, no H351) and IARC official live classification data (webapi.iarc.who.int/loc/loc.app.js, CAS 80-08-0 Dapsone Group 3 not-classifiable) shows neither cited source substantiates H351; original card...

First verdict *(2026-08-02T17:06, REFUTED)* live-receipt provenance audit vs pinned OSHA 29 CFR 1910.1000 Table Z-2 roster: toluene is claimed Table Z-1 in 3 signed receipts but is actually a Table Z-2 substance with an unscored ceiling/peak structure

Limits: no independent-review receipt; no movement event.

molecule-generator

PRE-PHASE-0 UNVALIDATED · 6 verdicts (5 refuted / 1 survived) · 1 KPIs, 5 repair receipts · queue: 2 pending / 3 done / 0 blocked

Most recent verdict *(2026-08-05T15:05, SURVIVED)* independent-verify V-A06v2: re-derived amine-centre and active-hydrogen functionality conventions for all 38 accepted proposals from raw canonical SMILES using a bond-type-enumeration RDKit implementation structurally distinct from both of R-A06v2 own implementations, validated instrument against...

First verdict *(2026-08-02T12:06, REFUTED)* percentile-bootstrap-CI-admissibility-at-k=n-boundary: reproduced target receipt bootstrap [1.0000,1.0000] over k=38/n=38, showed degeneracy is structural (45/45 seed x B x resample-size combos collapse, off-boundary...

Limits: no independent-review receipt; no movement event.

viscosity-score

PRE-PHASE-0 UNVALIDATED · 6 verdicts (3 refuted / 2 survived) · 4 KPIs, 2 repair receipts · queue: 2 pending / 1 done / 5 blocked

Most recent verdict *(2026-08-05T09:17, REFUTED)* Continued C9 open-literature sweep: closed 2 open leads (RSC Advances 10.1039/c5ra06914a, De Gruyter 10.3139/217.3338 – both confirmed closed-access via Unpaywall). Found a new, unambiguous CC BY 4.0, on-scope candidate (Finke et al. 2021, Materials 14(11):2752, PMC8196971, boehmite nanoparticles...

First verdict *(2026-08-01T21:11, SURVIVED)* sourced a real CC-BY 4.0 epoxy-coating HB dataset (Amorim et al 2021 SN Appl Sci 3:236), ran the module exact HB equation against published tau0/K/n at 4 shear rates, gated on the papers own Fig3 axis bounds + monotonic...

Limits: no independent-review receipt; no movement event.

cure-potlife-gate

PRE-PHASE-0 UNVALIDATED · 5 verdicts (4 refuted / 1 survived) · 1 KPIs, 1 repair receipts · queue: 1 pending / 1 done / 3 blocked

Most recent verdict *(2026-08-03T12:06, REFUTED)* Ea-free ablation: scored a universal chemistry-blind Q10=2 heuristic (ASTM F1980-16 default acceleration-factor convention, no literature Ea, no per-system fitting) against the signed per-system published-Ea Arrhenius model on the identical locked 36-point truth table using the KPI own metric

First verdict *(2026-08-01T01:09, SURVIVED)* ported the frozen Kissinger/OFW-Doyle/Starink CC-BY estimator+dataset (amphora-redteam-audit benchmarks) into amphora-cure-potlife-gate/benchmarks/kissinger-ozawa-baseline-v1/; head-to-head on the one exact chemistry...

Limits: no independent-review receipt; no movement event.

ingredient-library

PRE-PHASE-0 UNVALIDATED · 5 verdicts (2 refuted / 3 survived) · 1 KPIs, 3 repair receipts · queue: 1 pending / 2 done / 0 blocked

Most recent verdict *(2026-08-05T17:08, SURVIVED)* DSSE-verified source receipt signature + facts.json drift check + diagnostic-binding check + independent recompute of n_total/n_within_gate/per-row within_gate labels from the 9 raw comparison rows (not trusting stored summary/labels) + confirmed the 1 disclosed divergence is a pre-existing catalog...

First verdict *(2026-08-02T10:13, SURVIVED)* external live manufacturer-TDS re-verification of the 4 load-bearing hardener density facts (+2 AHEW, 1 amine-value byproduct) that had zero coverage in any of the 4 prior signed receipts; also independently tested...

Limits: no independent-review receipt; no movement event.

molecule-novelty

PRE-PHASE-0 UNVALIDATED · 5 verdicts (3 refuted / 2 survived) · 1 KPIs, 4 repair receipts · queue: 2 pending / 2 done / 0 blocked

Most recent verdict *(2026-08-05T04:07, SURVIVED)* independent fresh RDKit reimplementation (5th independent IPDA re-derivation, index-based LOO) of the four-family conformal novelty table on dedup-350 corpus; diffed all 20 cells vs shipped R-A07v2 table; independent DSSE verify of R-A07v2 against pinned trust-store key; chain byte-intactness...

First verdict *(2026-08-02T13:09, REFUTED)* corpus-duplicate-exchangeability audit: canonicalized every calibration-corpus PSMILES row and found only 350/443 rows are distinct structures (129 duplicate rows, 36 alias groups up to 12x replication e.g....

Limits: no independent-review receipt; no movement event.

molecule-role-ranker

PRE-PHASE-0 UNVALIDATED · 5 verdicts (1 refuted / 4 survived) · 2 KPIs, 4 repair receipts · queue: 3 pending / 2 done / 1 blocked

Most recent verdict *(2026-08-04T09:22, SURVIVED)* V-A12a independent verify of R-A12: from-scratch PAE reconstruction confirmed both original reinstated artifacts fail Ed25519 verification against the pinned evaluator key but pass against the local-dev key (defect real, not fabricated); confirmed R-A12 anchored batch receipt verifies against the...

First verdict *(2026-08-02T15:26, SURVIVED)* live-runtime generalization probe: real production RDKit C++ FFI (amphora-rdkit-ffi molecule_facts, pinned RDKit 2026.03.2) invoked on 10 real, cited, held-out coating-relevant molecules (cycloaliphatic diepoxide,...

Limits: no independent-review receipt; no movement event.

redteam-audit

PHASE-0 NOT-APPLICABLE · 5 verdicts (4 refuted / 1 survived) · 1 KPIs, 8 repair receipts · queue: 7 pending / 5 done / 5 blocked

Most recent verdict *(2026-08-05T12:09, REFUTED)* built a well-formed FALSIFICATION_ATTACK receipt against the real DGEBA EEW linear-series claim with a refutation criterion (predicted EEW<0/NaN/inf) that is algebraically incapable of firing for n>=0; signed with the real evaluator key; ran the actual unmodified lib/remote_verify.py...

First verdict *(2026-08-03T22:05, REFUTED)* portfolio-wide retraction-index completeness sweep: enumerated 15 modules with >=1 REFUTED ledger verdict from a pinned sha256-snapshotted ledger, checked each for evidence/phase0/kpis/RETRACTION-INDEX.md presence and...

Limits: evaluator and reviewer receipts present, movement event recorded, work queued.

design-rule-score

PRE-PHASE-0 UNVALIDATED · 4 verdicts (3 refuted / 1 survived) · 3 KPIs, 1 repair receipts · queue: 1 pending / 1 done / 2 blocked

Most recent verdict *(2026-08-02T21:05, REFUTED)* leak-free re-run of forward-chained prospective-transfer benchmark over the DEDUP first-appearance partition established by split-integrity-dedup-v1, removing the 27/42 (64%) duplicate test rows memorized from train in the train{1,2}->test{3} fold

First verdict *(2026-08-01T02:05, REFUTED)* Propagate split-integrity-dedup-v1 leak-free re-baseline (2026-07-30) into module headline KPI: write superseding KPI receipt recording HistGBM leak-free collapse (R2=-0.0469 vs constant-mean -0.0535), Guth-Gold physics...

Limits: no independent-review receipt; no movement event.

ip-fto-guard

PRE-PHASE-0 UNVALIDATED · 4 verdicts (4 refuted / 0 survived) · 1 KPIs, 2 repair receipts · queue: 3 pending / 1 done / 0 blocked

Most recent verdict *(2026-08-05T07:10, REFUTED)* claim-construction sensitivity ladder (narrow/plain/broad-spec-informed) over the 6 claim-charted families (patent_landscape.json independent_claim_chart.charted=true), re-deriving real claim-1 text + specification disclosure from the already hash-pinned Google Patents snapshots; family n=26...

First verdict *(2026-08-02T11:16, REFUTED)* schema-vs-live-runtime metric parity audit: static extraction of evaluate_ip_fto_guard() metrics keys + repo-wide grep + checked-KPI-test-coverage scan + signed-benchmark-artifact scan for the declared primary metric key

Limits: no independent-review receipt; no movement event.

tolerance-robustness

PHASE-0 LOCKED-INTERNAL · 3 verdicts (3 refuted / 0 survived) · 5 KPIs, 1 repair receipts · queue: 1 pending / 2 done / 1 blocked

Most recent verdict *(2026-08-05T05:09, REFUTED)* Repair: re-verify V1-RESOLVE two NOT_RESOLVED rows (literal PENDING; mismatched label-admissibility referent) by rewriting both superseded-by cells to honestly disclose no repair exists and name campaign item R-C01, then re-ran the original unmodified V1-RESOLVE mechanical sweep script against the...

First verdict *(2026-08-02T06:11, REFUTED)* Re-derived RadonPy PI1070 eps_r split-conformal coverage restricted to physically-admissible subset (eps_static>=n^2, Frohlich/Debye); ran scaffold-safe (non-exchangeable) split-conformal on OpenPoly Tg

Limits: evaluator and reviewer receipts present, movement event recorded, work queued.

assurance-capabilities

? ? · 2 verdicts (1 refuted / 1 survived) · 0 KPIs, 3 repair receipts · queue: 1 pending / 2 done / 0 blocked

Most recent verdict *(2026-08-05T16:23, SURVIVED)* repair W-ASSUR1: fixed normalized_key_skeleton NFKC-recompose bug (switched to NFKD) so U+00FC and other precomposed-diacritic homoglyphs fold to ASCII instead of being dropped; re-ran V-B01 5 novel probes verbatim + original A1..A8 (13/13 detected, two-replay byte-identical); positive control...

First verdict *(2026-08-04T23:15, REFUTED)* 5 novel candidate JSON encodings (uncovered-homoglyph letter-deletion x2, ZWJ-in-key, deeply-nested array-of-objects alias, fullwidth-unicode-escape key) via standalone cargo bin against real...

Limits: no independent-review receipt; no movement event.

known-answer-benchmarks

PHASE-0 LOCKED-INTERNAL · 2 verdicts (1 refuted / 1 survived) · 0 KPIs, 2 repair receipts · queue: 1 pending / 1 done / 0 blocked

Most recent verdict *(2026-08-04T05:08, SURVIVED)* independent verify of R-A05: fresh from-scratch exact-rational re-derivation of both volume-solids controls (not R-A05 own script), live re-fetch of A221 TDS to cross-check transcription, independent enumeration of all 14 raw controls for sweep completeness, Ed25519 re-verify of R-A05 DSSE against…

First verdict *(2026-08-03T07:08, REFUTED)* independent exact-rational (Fraction) re-derivation of Mix volume-solids% for the A221-X70P1 primer from raw ingredient masses + component density/solids-percent numbers transcribed from the same cited Evonik…

Limits: evaluator and reviewer receipts present, movement event recorded, work queued.

process-qc-twin

PHASE-0 LOCKED-INTERNAL · 2 verdicts (1 refuted / 1 survived) · 6 KPIs, 1 repair receipts · queue: 2 pending / 1 done / 2 blocked

Most recent verdict *(2026-08-05T03:05, SURVIVED)* Resumable re-file: verified pre-staged two-replay bytes identical, ran pre-staged signer for the MRbar/d2 x EWMA/CUSUM memory-detector ARL cross, self-verified DSSE, then superseded the module consolidated ARL-characterization KPI (v1->v2) to fold the result in and updated mission.md +…

First verdict *(2026-08-04T16:21, REFUTED)* Promoted 4 signed ARL-degradation audits into a consolidated Phase-0 KPI (evidence/phase0/kpis/); launched the MRbar/d2 x EWMA/CUSUM cross-check run (attacks estparam-memory-detector-arl-degradation-v1 m=50…

Limits: evaluator and reviewer receipts present, movement event recorded, work queued.

rdkit-ffi

? ? · 2 verdicts (1 refuted / 1 survived) · 0 KPIs, 3 repair receipts · queue: 1 pending / 2 done / 0 blocked

Most recent verdict *(2026-08-05T18:16, REFUTED)* Fresh independent Rust FFI harness (ffi1-verify, real production molecule_facts, not reimplemented) validated 14 pre-registered probes across 10 amine classes against externally-published active-hydrogen counts (Lee & Neville 1967 convention; vendor TDS/CAS MW+formula cross-checked; patent…

First verdict *(2026-08-04T03:19, SURVIVED)* V-N02c independent verify: re-ran cargo test -p amphora-rdkit-ffi –features rdkit (7/7 green incl. R-N02 regression test); direct source read of bridge.cc confirmed is_amidine_or_guanidine_carbon is structurally…

Limits: no independent-review receipt; no movement event.

candidate-contract

PHASE-0 NOT-APPLICABLE · 1 verdicts (0 refuted / 1 survived) · 1 KPIs, 1 repair receipts · queue: 2 pending / 1 done / 0 blocked

Most recent verdict *(2026-08-05T19:23, SURVIVED)* 8-vector adversarial-encoding injection (top-level key, array container, deep nesting, key-case variant, alternate key name, semantic-alt field, string container, unicode-homoglyph key) of candidate-supplied MW/EEW/AHEW/group_count/canonical-identity, run against the REAL compiled amphora-engine…

Limits: a single verdict is not coverage; only survivals so far — nothing has yet been found wrong here, which is weaker evidence than something having been.

cro-protocol-packager

PHASE-0 NOT-APPLICABLE · 1 verdicts (1 refuted / 0 survived) · 1 KPIs, 1 repair receipts · queue: 2 pending / 1 done / 0 blocked

Most recent verdict *(2026-08-05T20:13, REFUTED)* directive T2-02 (a-d) + 8-alternate-encoding injection harness (16 vector/path combos) against CroProtocolPackageV2 claim_locks and execution_authorized live runtime

Limits: a single verdict is not coverage.

envelope-gates

PHASE-0 NOT-APPLICABLE · 1 verdicts (1 refuted / 0 survived) · 1 KPIs, 1 repair receipts · queue: 1 pending / 1 done / 1 blocked

Most recent verdict *(2026-08-05T21:11, REFUTED)* metamorphic boundary-fuzz grid (4 classes x 4 fail-closed metrics, 18 cells) vs evaluate_gate() in run_envelope_gates.py: inclusive-bound off-by-one 1ULP, inverted envelope, NaN/Inf bounds, unit-scale substitution; excursion-invariance/value-free-bypass probe

Limits: a single verdict is not coverage.

evidence-intake-validator

PHASE-0 NOT-APPLICABLE · 1 verdicts (1 refuted / 0 survived) · 1 KPIs, 1 repair receipts · queue: 2 pending / 1 done / 1 blocked

Most recent verdict *(2026-08-04T20:16, REFUTED)* honest-host real Ed25519 verification + real trust-store lookup over n=66 real manifest receipts + n=9 known-bad R-A11/R-A12 receipts, fed into verbatim host_receipt_valid re-derivation

Limits: a single verdict is not coverage.

model-card-authority

PHASE-0 NOT-APPLICABLE · 1 verdicts (0 refuted / 1 survived) · 2 KPIs, 1 repair receipts · queue: 2 pending / 1 done / 0 blocked

Most recent verdict *(2026-08-04T13:09, SURVIVED)* construct KPI from real audit counts (no invented n) + re-execute live differential probe today as currency falsification probe

Limits: a single verdict is not coverage; only survivals so far — nothing has yet been found wrong here, which is weaker evidence than something having been.

source-ledger

PHASE-0 NOT-APPLICABLE · 1 verdicts (0 refuted / 1 survived) · 1 KPIs, 1 repair receipts · queue: 2 pending / 1 done / 0 blocked

Most recent verdict *(2026-08-04T21:08, SURVIVED)* independent non-runtime re-derivation of citation-coverage predicate directly against REAL amphora-ingredient-library/facts.json (byte-identical mirror check, per-key gap enumeration, named recent-addition resolvability, byte-tamper probe, cross-check vs 2 prior signed oracles)

Limits: a single verdict is not coverage; only survivals so far — nothing has yet been found wrong here, which is weaker evidence than something having been.

LIFT VS BEST KNOWN PRACTICE

What is genuinely ahead, and what is not

AHEAD OF COMMON PRACTICE

- **Refutation is first-class.** 68 of 108 verdicts are the system publishing its own defects. Standard practice publishes wins and files failures in a tracker; here a refutation is a signed, append-only artifact that cannot be quietly withdrawn.
- **Every claim is content-addressed.** A verdict names the artifact it attacks by SHA-256, and the launcher re-derives attribution from a pre-fire content snapshot rather than trusting file timestamps.
- **Survivals must carry a positive control.** A survival is only credited if the fire also planted a defect of the class its attack exists to catch and showed the attack fired on it. Most benchmark suites never demonstrate that a passing test could fail.
- **The audit trail includes the auditor's own failures** — including, this edition, a commit message that claimed a guard which did not exist. That correction is a committed file, not a deleted line.

BEHIND, AND BY HOW MUCH

- **No owned wet-lab measurement exists anywhere in the portfolio.** Every number rests on public corpora, literature values or simulation. Industrial coating development is measured on panels; this portfolio has never measured a panel.
- **Zero modules at Phase-1.** Independent higher-phase validation has not been attempted, because it needs the measurement above.
- **Sample sizes remain small where it matters.** The design-rule constraint is 58 unique observations from one formulator; the barrier literature series are n=3–5, below the module's own feasibility floor of 6.
- **Applicability domain is narrow and honestly so.** Four of five canonical industrial amines fall outside the property models' domain at the current threshold — the gate abstains rather than guessing, which is correct behaviour and a real limit.

Red team. The lift claimed above is a lift in *method*, not in predictive accuracy. Nothing in this document demonstrates that Amphora predicts a coating property better than an incumbent method on owned data, because no owned data exists. A reader who takes 'ahead of common practice' to mean 'better predictions' has been misled — and this paragraph exists so that reading is not available.

ERRATA

Corrections against the 2026-08-04 edition

NUMBERS THAT CHANGED, AND WHY

quantity	08-04	08-05	why
ledger verdicts	78	108	30 fires landed verdicts in the interval
autonomous fires	209	238	one hour lost to a launcher defect (see ops doc)
signed receipts	444	193	counted here as 75 repair + 118 audit DSSE, a narrower and more honest basis
campaign items	121	152	directive obligations filed as tracked items

A CLAIM WITHDRAWN DURING PREPARATION OF THIS DOCUMENT

Withdrawn before publication. While preparing this edition an analysis concluded that the `PHASE0_INTERNAL_REVIEWER` key had *never been used*, and that Phase-1 was therefore structurally unreachable. **That was wrong.** The analysis scanned only repair receipts, where the absence of a reviewer signature is correct, and generalised it to the whole tree. Re-checking every DSSE block found 18 reviewer signatures across 18 modules, and all 9 derived-Phase-0 modules carrying both roles plus a movement event. The claim was caught by a verification step that existed specifically because it was going to be the headline. It is recorded here rather than quietly dropped.

STANDING DATA DEFECTS

- 2 ledger rows carry a stray leading backslash on their timestamp, from a shell-interpolation bug now fixed at source. The ledger is append-only, so the rows are permanent; every consumer must strip the prefix. Repairing them would destroy the tamper-evidence property that makes the ledger worth having.
- The Phase-0 decision report is regenerated by an external command, not by this pipeline, so it lags. This edition quotes it dated.

WHAT NEEDS A HUMAN

Decisions the loop is forbidden from taking

The loop files and tracks these; it can never close them. A guard in the scheduler refuses to dispatch a fire against any of them.

ORDERED BY WHAT THEY UNBLOCK

1. **Register two trusted signer identities.** Distinct, non-revoked, non-development Ed25519 keys for evaluator and reviewer. Gates the re-evaluation ceremony and every signed lab return. *Also adopt the triage policy: do not mass re-sign the pre-loop receipts — that would make the chain look clean while destroying the distinction that makes it tamper-evident.*
2. **Run the re-evaluation ceremony.** Restores 7 of the 8 gap modules. The tool is built and tested and has never been run. `molecule-property` is not restorable this way.
3. **Send the six rights and data asks.** Each gates one wet-lab experiment. The USBR request is written, signed and waiting on a human to send it.
4. **Contract the eight independent counterparties.** The two signer roles must be distinct from each other and from the lab, or a Phase-1 claim means nothing.
5. **Decide the formulation-universe merge** — held pending the first attack on that module, which has now been queued.

The sequencing fact that governs everything. Nothing crosses the Phase-1 line until at least one experiment returns a signed, candidate-bound measurement passing its locked gate under two distinct trusted signatures. Until then, derived Phase-0 is a provenance statement, not a scientific one, and the honest external number stays at 9 modules and zero at Phase-1. The lab is not an accelerant on the current trajectory; it is the only thing that changes it.

HOW TO VERIFY ANYTHING IN THIS DOCUMENT

Every figure derives from `data/facts.json`, which is computed from a read-only snapshot of the evidence tree. The snapshot is hash-pinned in `data/.FRESH.sha256`. To re-derive: run `scripts/01_pull_data.sh` then `scripts/02_analyze.py` and diff. No number in this document was typed by hand.